

Irony, not paradoxical!

Gödel's incompleteness theorems in a new key

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Abstract

The presentation offers a reinterpretation of Gödel's results on the incompleteness of mathematics in the framework of his general position in philosophy of logic. The guiding reading key of the proposal is an 'ironic' one, which defends that Gödel assumes the philosophical presuppositions common to all logical formalisms of his time in order to take this 'game' to the limit of its consequences. In other words, it is a *reductio ad absurdum*. The main philosophical presuppositions discussed are the notion of proposition and the use of predicates in formal systems. By revisiting texts prior and after Gödel's proofs, we will examine Vienna's logician stance on these presuppositions and, in the light of this, make explicit what is it that Gödel is doing both in his proofs of completeness of first-order logic and incompleteness of second-order logic. It is concluded that what we have perceived as a paradox is rather an argumentative strategy challenging the philosophical basis of formal systems, aligning his project with Frege's approach to logic and mathematics, rather than foundational projects of mathematics and model theory.

References

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